

Crop Price Prediction: The Freezer Gambit Solution

Presented by Team Xforce



Problem Understanding

Core Challenge

Predict crop prices 4 weeks ahead in Agrovía

Business Goal

Maximize profit for "Freezer Gambit" strategy

Key Decision

When to sell fresh vs. freeze produce weekly





Data Overview & Preparation

Price Data

2040-2043,
includes Date,
Region, Commodity,
Price

Weather Data

Temperature,
Rainfall; duplicates
handled via merging

Preparation

Renaming, type
conversion, data
merging for
modeling

Feature Engineering Strategy

Time-Based Features

Year, month, week, day_of_week

Capture seasonality

Price Lag Features

Lags at 28, 35, 42 days

Capture autocorrelation

Rolling Window Features

Mean, std over 7, 14, 28 days

Capture recent trends/volatility

Model Selection & Training

1 Model Choice

LightGBM Regressor for performance and speed

2 Data Handling

Supports categorical features well

3 Validation

Time-based split: train until 2043-04-30, validate May-June 2043



Hyperparameter Tuning & Performance

Optuna Tuning

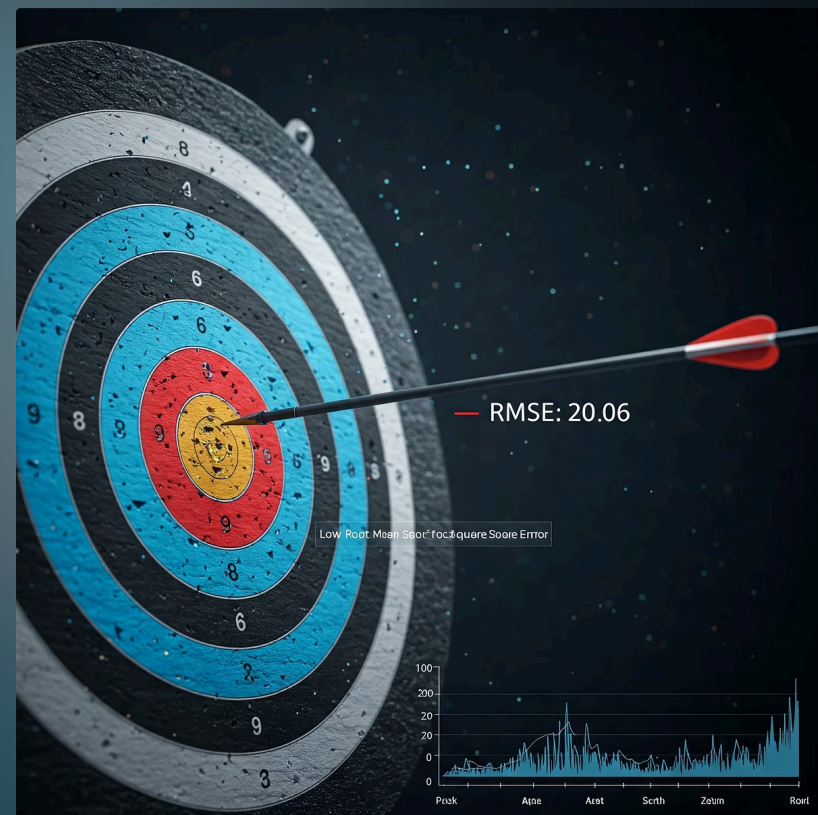
30 trials optimizing LightGBM parameters

Performance Metric

Final Validation RMSE: **20.06**

Significance

Low RMSE signals accurate unseen data predictions



Feature Importance Insights



Commodity

Different crop types affect price prediction.



Region

Location significantly influences crop prices.



Price Lag

Past prices at 28, 35, and 42 days help predict future trends.



Price Volatility

Recent price trends and fluctuations improve prediction accuracy.

This confirms the critical role of location, crop type, past prices, and price volatility for precise predictions.

System Architecture & API



Pipeline

Data → Preprocessing →
Feature Engineering →
Training → Saved Model



API Delivery

FastAPI app containerized
with Docker



Endpoint

Key `/api/predict` for crop price forecasts





Business Insights & Recommendations



Insights

Price driven by commodity, region, and recent price moves



Application

Supports "Sell vs. Freeze" decision-making



Next Steps

- Integrate cost data for better decisions
- Develop decision support tool
- Implement data ingestion for placeholder APIs
- Continuous model performance monitoring

Conclusion & Thank You

High-accuracy LightGBM forecasting model delivered (RMSE 20.06)

Dockerized API meets competition requirements

Future work: expanded features, integration, ongoing validation

Thank You!

Questions?